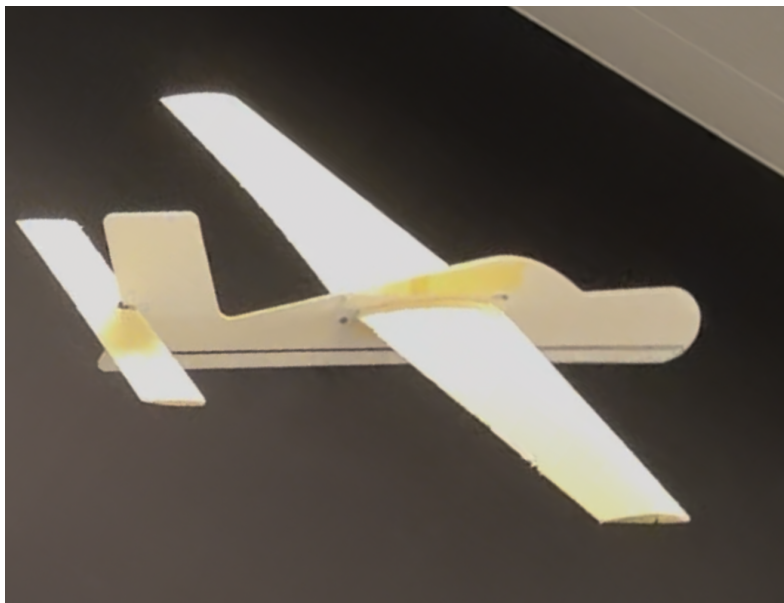




EPFL
Semester Project Report

Codesign of Shape and Control: A Study in Autonomous Perching

Semester Project (Master)



Glider prototype for the project.

Program Master in Robotics
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Year 2025

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1 Introduction

This semester project focuses on the design and prototyping of a small, hand-launched, unpowered fixed-wing glider intended to perform autonomous perching maneuvers, in the spirit of the work by Moore et al. [2014]. In this context, perching refers to a controlled terminal maneuver in which the glider transitions from a stable glide to a high-angle-of-attack, low-speed approach in order to make contact with (and ideally stop on) a designated perching location in a repeatable manner.

Compared to the reference perching platform, our prototype is intentionally modified to serve as an experimental platform for evaluating shape-optimisation algorithms, with a particular emphasis on the airfoil geometry. The current iteration uses an asymmetric airfoil, selected as a practical and performance-oriented baseline. This prototype is therefore intended to act as a reference design on top of which future work can systematically optimise the wing shape (airfoil and potentially other geometric parameters), while maintaining a consistent experimental setup for meaningful comparisons.

To support rapid iteration and fair comparisons between candidate shapes, the mechanical design places strong emphasis on modular wings, enabling fast swapping of wing configurations while keeping the rest of the aircraft and control architecture as constant as possible.

At a high level, the onboard system is designed to provide the actuation, power, and communication required for perching experiments. The control architecture relies on two microcontrollers (a main controller and a secondary one for dedicated tasks), and specifically actuates the elevator via a servo motor to control the aircraft's pitch. A lightweight radio link is used for transmitting commands during development and testing. Power is supplied by a 2S Li-Po battery regulated to 5 V, and basic safety monitoring is implemented through battery-voltage measurement to prevent deep discharge during repeated experimental runs.

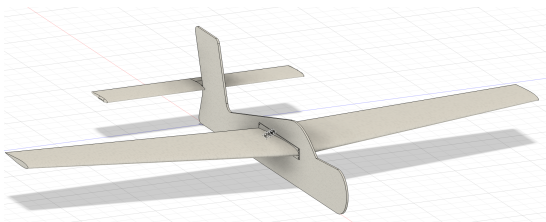


Figure 1: glider design

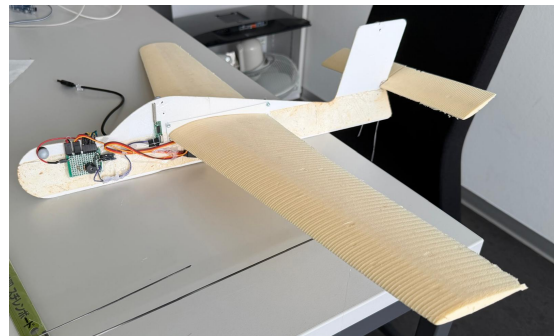


Figure 2: glider implementation

2 Flight Objectives and Assumptions

This project aims to design a mini glider (hand-launched, unpowered) inspired by the platform used by Moore et al. [2014]. While the reference glider weighed approximately 100 g with simple flat-plate wings, we established a target mass budget of 150 g. This 50% increase was chosen to accommodate the more complex construction of a modular, cambered wing (as opposed to a simple flat plate) and to provide a safety margin for the onboard electronics and mechanics. In practice, the final prototype utilized this entire budget, weighing exactly 150 g.

To ensure the glider remains hand-launchable at a moderate speed (≈ 7 m/s) despite the increased mass, we determined the required wing area S analytically. We target stable flight at this low speed to facilitate the transition to perching maneuvers. These baseline flight conditions establish the operational envelope for subsequent perching maneuvers, where the glider will transition to high angles of attack.

To estimate the required lift, we use the aerodynamic lift formula:

$$L = \frac{1}{2} \rho V^2 S C_L \quad (1)$$

where L is lift, ρ is air density, V is speed, S is wing area, and C_L is the lift coefficient [Wikipedia contributors]. In steady, level flight, L must equal the aircraft weight (mg). Choosing a moderate lift coefficient $C_L = 0.5$ (typical at moderate angle of attack, well below stall), we size the wing area so that $L = mg$. Taking $\rho \approx 1.225$ kg/m³ (standard air), we get:

- Weight: $mg \approx 1.47$ N for $m = 0.15$ kg.
- Required wing area:

$$S = \frac{2L}{\rho V^2 C_L} \approx \frac{2 \times 1.47}{1.225 \times 7^2 \times 0.5} \approx 0.10 \text{ m}^2.$$

We therefore adopt a wing area $S \approx 0.10$ m² to guarantee the needed lift at 7 m/s.

To validate our airfoil selection and lift assumptions, we must first determine the flow regime. Assuming a typical aspect ratio of $AR \approx 7$ for a glider of this size, the mean chord length is approximately $c = \sqrt{S/AR} \approx 0.12$ m. With a flight speed $V \approx 7$ m/s and air viscosity $\mu \approx 1.81 \times 10^{-5}$ Pa·s, the Reynolds number is:

$$Re = \frac{\rho V c}{\mu} \approx \frac{1.225 \cdot 7 \cdot 0.12}{1.81 \times 10^{-5}} \approx 5.7 \times 10^4.$$

This value places the aircraft in the **low-Reynolds-number regime**. Consequently, we must rely on "low- Re polars"—aerodynamic performance graphs (such as C_L vs. α) specifically generated for this viscous flow regime—rather than standard high-Reynolds data used in general aviation.

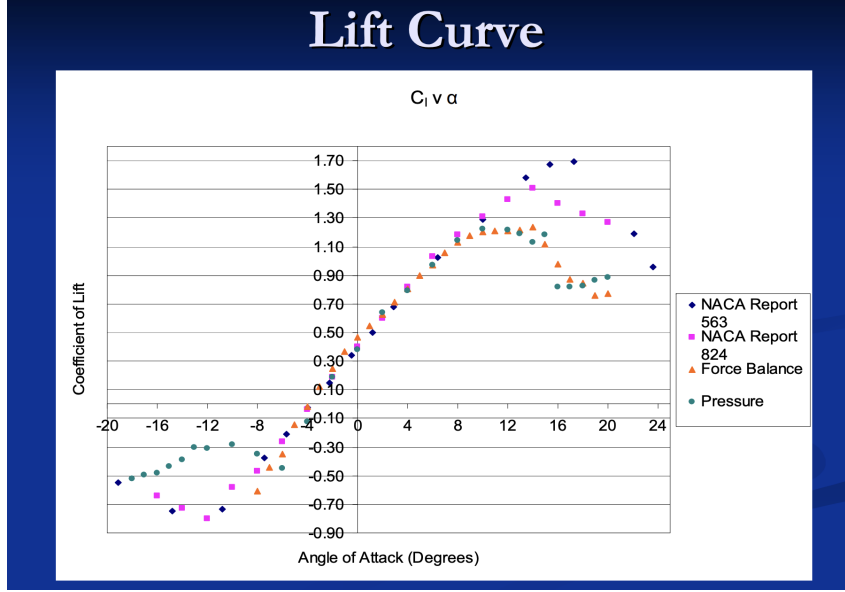


Figure 3: Lift Curve for NACA 4412 airfoil Baylor University.

For the following calculations, we assume $C_L = 0.5$, which is a reasonable approximation for stable, non-stalled flight at small angles of attack. As shown in the Figure “Lift Curve”, the $C_L(\alpha)$ relationship for the NACA 4412 is reported and compared against published NACA datasets (NACA Report 563 and 824). The inclusion of these historical reports serves to validate the aerodynamic behavior of the airfoil across different experimental conditions and Reynolds numbers. Despite variations in testing setups, a consistent trend emerges across these separate studies: the target lift coefficient of $C_L = 0.5$ is reliably achieved at a very low angle of attack, typically between 0° and 4° .

Low- Re polars for the NACA 4412 ($Re \approx 5.5 \times 10^4$) further confirm this, showing that $C_L \approx 0.48$ at $\alpha = 2^\circ$ and $C_L \approx 0.53$ at $\alpha = 2.5^\circ$, meaning $C_L = 0.5$ occurs around $\alpha \approx 2.2^\circ$ (Figure “Lift Curve”). We adopt this value for simplicity in our calculations. This operating point remains comfortably below the stall region and provides margin for disturbances and minor variations in flight conditions.

2.1 Wing Geometry

To design an efficient wing, we choose an aspect ratio $AR = 7$. The aspect ratio ($AR = \frac{b^2}{S}$, with b the span) measures wing slenderness (span-to-chord). An $AR = 7$ is a suitable compromise for a small glider: high enough to improve glide ratio (reducing induced drag) without overly stressing wing rigidity. For hand-launched gliders, an AR between roughly 4.5 and 7.5 is typically recommended to avoid overly slender wings breaking on launch gli. Our choice $AR = 7$ sits at the top of this practical range, reducing induced drag efficiency for a small glider while remaining buildable and strong.

Knowing S and AR , we compute span b and mean chord c from $b = \sqrt{AR \cdot S}$ and $c = \frac{S}{b}$. With $S = 0.10 \text{ m}^2$ and $AR = 7$:

- Span: $b \approx \sqrt{7 \times 0.10} \approx 0.84 \text{ m}$ (about 84 cm).

- Mean chord: $c \approx \frac{0.10}{0.84} \approx 0.12$ m (about 12 cm).

These dimensions confirm a compact airframe (mini glider) consistent with the 0.15 kg payload target.

Wing planform: We adopt a trapezoidal wing (straight, unswept) with a moderate taper ratio $\lambda = 0.6$ (tip-chord to root-chord ratio). This means the chord at the root (fuselage junction) will be approximately 1.67 times larger than the chord at the tip (since $0.6 = c_{\text{tip}}/c_{\text{root}}$). According to Prandtl’s lifting-line theory, an elliptical lift distribution minimizes induced drag Anderson [2010]. While an elliptical planform achieves this naturally, a trapezoidal wing with a suitable taper ratio can closely approximate this distribution, offering a significant aerodynamic advantage over a rectangular wing while being easier to manufacture than a purely elliptical one. For comparison, the theoretical optimal taper ratio for minimizing induced drag on an unswept wing is approximately 0.45 Aviation Stack Exchange. Our choice of 0.6 favors a moderately tapered wing, avoiding an overly thin tip (which would be structurally fragile and prone to local stall Raymer [2018]) while reducing weight compared to a rectangular wing. This compromise provides good aerodynamic efficiency without complicating construction.

Wing dihedral: The wing will incorporate a positive dihedral angle of 2° (angle between each semi-wing and the horizontal plane). Dihedral improves lateral stability: when a roll disturbance affects the aircraft, the dihedral creates an automatic restoring moment gli. Typically, gliders without ailerons often have around 5° of dihedral to ensure sufficient roll stability gli. Here, with only 2° , the dihedral is lower, likely assuming the presence of lateral control (ailerons) or a reduced need for passive stability. This slight 2° dihedral will still provide natural roll stability while limiting the reduction in lift induced by excessive dihedral.

Wing incidence: The wing will be set at $+2^\circ$ incidence relative to the fuselage axis (elevator set at 0°). This longitudinal decalage (wing-tail incidence difference) of 2° provides a natural pitch-up moment: in stabilized flight, the fuselage will be nearly aligned with the airflow, with the wing generating slightly more lift than the empennage. Typical incidence settings in light aviation and model aircraft fall within this range: for example, the wing is often set between 0° and $+5^\circ$, and the elevator between $+2^\circ$ and -2° , yielding a longitudinal decalage of approximately 0° to 4° amcco. Our choice ($\approx 2^\circ$) is therefore standard practice to ensure longitudinal stability: the glider will tend to return to its equilibrium angle of attack after a disturbance, without pitching up or diving violently.

2.2 Airfoil Selection

The NACA 4412 airfoil was selected as a practical baseline for this prototype. While Moore et al. [2014] utilized a simple flat-plate wing for their perching glider, we intentionally chose a cambered airfoil to introduce a more representative aerodynamic geometry and to explore a more complex manufacturing process (CNC machining of 3D shapes). The NACA 4412 is a classic "trainer" airfoil known for its gentle stall characteristics and good lift production at low Reynolds numbers ($Re \approx 5.7 \times 10^4$), making it a suitable can-

didate for a stable experimental platform. Furthermore, cambered profiles are commonly used on gliders and can still support aerobatic-capable designs. For instance, Dodgson [2023] describes an RC glider design that was "fully aerobatic" and used a NACA 4412 airfoil on one of its wing sets. This choice allows us to validate the manufacturing and flight testing workflow on a standard asymmetric profile before potentially optimizing the shape in future iterations.

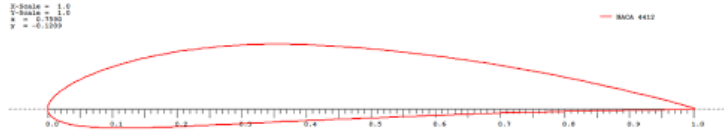


Figure 4: NACA 4412 airfoil profile.

2.3 Horizontal Tail (Stabilizer)

To ensure pitch stability and control, we design a conventional horizontal tail. An empirical design rule states that the stabilizer surface should represent approximately 20% of the wing area Scribd. We have adopted this approach: $S_h \approx 0.2 S$, which yields a horizontal tail area $S_h \approx 0.2 \times 0.10 = 0.020 \text{ m}^2$. This value corresponds to what is observed in many light aircraft: for example, the stabilizer of an aircraft typically represents $\sim 20\%$ of the wing area, and the vertical fin $\sim 10\%$ dri. A stabilizer representing $1/5$ of the wing area generally provides sufficient restoring moment without adding excessive drag or weight.

Moment arm and tail volume: The horizontal tail volume ratio (tail volume coefficient V_H) is a key parameter for longitudinal stability. It is defined by:

$$V_H = \frac{S_h \cdot l_h}{S \cdot c}$$

where S_h is the horizontal tail area, S is the wing area, c is the mean wing chord, and l_h is the horizontal tail moment arm. In this report, l_h is defined as the distance between the wing aerodynamic center and the horizontal tail aerodynamic center. This convention is used consistently throughout the sizing and stability calculations.

We have targeted a volume coefficient $V_H \approx 0.5$, a typical value for small stable aircraft. Indeed, gliders or light aircraft typically have V_H in the range 0.5–0.7 vib (for example ~ 0.6 for a glider). Setting $V_H = 0.5$ allows us to calculate the required moment arm:

$$l_h = \frac{V_H \cdot S \cdot c}{S_h} \approx \frac{0.5 \times 0.10 \text{ m}^2 \times 0.12 \text{ m}}{0.020 \text{ m}^2} \approx 0.30 \text{ m}.$$

We thus obtain a distance of approximately 30 cm between the wing (center of gravity) and the horizontal tail. This length typically corresponds to the rear section of a fuselage 50–60 cm long (accounting for the nose). Such a moment arm ensures good tail authority: the tail will be able to generate the necessary longitudinal moment to stabilize the aircraft or to pitch up/down in flight.

Horizontal tail dimensions: With $S_h \approx 0.02 \text{ m}^2$, we must choose its span b_h and mean chord c_h . For construction simplicity, we can maintain a rectangular or slightly tapered planform. The initial design suggested that the stabilizer chord be taken as half the main wing chord. If the wing chord is $\sim 12 \text{ cm}$, we would have a tail chord $c_h \approx 6 \text{ cm}$. Then the stabilizer span would be $b_h = S_h/c_h \approx 0.02/0.06 \approx 0.33 \text{ m}$, or 33 cm of stabilizer span. These orders of magnitude (6 cm chord, $\sim 33 \text{ cm}$ span) are consistent with a sufficiently elongated tail to be effective, without exceeding fuselage size constraints. Note that the horizontal tail airfoil will likely be symmetric. to generate stabilizing moments tai in both positive and negative directions without parasitic lift at neutral position.

2.4 Vertical stabilizer

The Vertical stabilizer ensures yaw stability and directional control. As with the stabilizer, we can size the fin area as a fraction of the wing area. Conventional aircraft typically have a fin corresponding to $\sim 10\text{--}15\%$ of the wing area tai. We have chosen $S_v \approx 0.10 S$, approximately 0.010 m^2 . This value represents 50% of the stabilizer area (0.5×0.02), which corresponds to a commonly observed fin-to-stabilizer ratio (the fin is about half the stabilizer in many designs). For example, the sesquiplane Blériot-SPAD S.27 had a horizontal tail of 2 m^2 (20% of the wing) and a Vertical stabilizer of 1 m^2 (10% of the wing) dri – similar to our proportions. A fin representing $\sim 10\%$ of the wing area generally provides adequate directional stability without adding excessive lateral drag.

Fin shape: We have chosen a trapezoidal fin with a more pronounced taper – the leading edge being longer than the trailing edge. A trapezoidal (tapered) vertical tail is commonly adopted to reduce structural loads at the root and structural weight; typical recommendations for conventional transport aircraft place the vertical-tail taper ratio around $\lambda_v \approx 0.3\text{--}0.6$ tai. Therefore, we select $\lambda_v \approx 0.5$ ($c_{\text{tip}} \approx 0.5 c_{\text{root}}$), meaning the chord at the fin base will be approximately twice the chord at the tip. This choice is motivated by both structural and aerodynamic considerations: the wider base provides more surface area near the fuselage (a region disturbed by it), and offers a robust attachment to the fuselage, while the taper reduces weight at height and bending moments on the fin. Moreover, concentrating the area toward the bottom improves directional efficiency (as the flow behind the fuselage benefits more from the lower portion of the fin). Many aircraft use a fin that is wider at the root than at the tip for these reasons.

Approximate fin dimensions: Assuming $S_v \approx 0.01 \text{ m}^2$, the dimensions could be, for example: a height (vertical span) h_v of approximately 15 cm, and a mean chord $c_{v,\text{mean}} \approx \frac{S_v}{h_v}$. If $h_v = 0.15 \text{ m}$, then $c_{v,\text{mean}} \approx 0.01/0.15 \approx 0.067 \text{ m}$ (6.7 cm). With $\lambda_v = 0.5$, we would have a root chord $c_{\text{root},v} \approx 9\text{--}10 \text{ cm}$ and a tip chord $c_{\text{tip},v} \approx 4.5\text{--}5 \text{ cm}$, which corresponds to the scale of the stabilizer (stabilizer chord $\sim 6 \text{ cm}$). These values remain indicative; the important point is that the fin provides sufficient area and yaw moment.

2.5 General Configuration

In summary, the preliminary characteristics of the glider are:

- **Expected mass:** ~ 150 g (small-scale mini glider).
- **Wing area:** ~ 0.10 m² (sized for sustentation at 7 m/s, $C_L = 0.5$).
- **Wing span:** ~ 0.84 m; **Mean wing chord:** ~ 0.12 m; **Aspect ratio:** $AR \approx 7$.
- **Wing planform:** Trapezoidal, taper ratio $\lambda \approx 0.6$; dihedral 2° for lateral stability; incidence angle $+2^\circ$.
- **Horizontal tail:** ~ 0.020 m² ($\approx 20\%$ of S); typical chord ~ 6 cm; span ~ 30 – 35 cm; tail volume $V_H \approx 0.5$ (moment arm ~ 30 cm).
- **Vertical stabilizer:** ~ 0.010 m² ($\approx 10\%$ of S); height ~ 15 cm; root chord ~ 9 cm, tip chord ~ 4 – 5 cm ($\lambda \approx 0.5$); tail volume $V_V \approx 0.03$ – 0.04 .
- **Airfoil:** Wing: NACA 4412 cambered profile favoring lift at low Reynolds number; tail: symmetric profile (stabilizer) and thin (fin).

3 Construction

3.1 Mass and Center of Gravity Constraints

The main challenge in designing the glider was to meet the target mass (150 g) while controlling the position of the center of gravity (CG). To ensure sufficient longitudinal stability, the CG must remain in the forward region of the mean aerodynamic chord (MAC). In particular, the Pilot's Handbook of Aeronautical Knowledge (FAA) indicates that designers place the CG "as close as possible" to the 20% MAC point (measured from the leading edge), and specifies that the aft CG limit is determined "largely by stability considerations" Federal Aviation Administration.

Aerodynamic center (cambered airfoil): For a cambered airfoil, the aerodynamic center is not exactly at $0.25c$. Using low- Re polars for the NACA 4412, we estimate

$$\frac{x_{ac}}{c} = 0.25 - \frac{C_{m0}}{C_{L\alpha}}.$$

With $C_{m0} \approx -0.0739$ and $C_{L\alpha} \approx 7.895 \text{ rad}^{-1}$, we obtain

$$\frac{x_{ac}}{c} \approx 0.259.$$

This reference point is used in the stability discussion.

Stability requirement: For longitudinal static stability, the center of gravity must be positioned *forward* (ahead) of the aerodynamic center (center of lift). The aerodynamic center, located at approximately 25.9% of the chord as calculated above, is the point

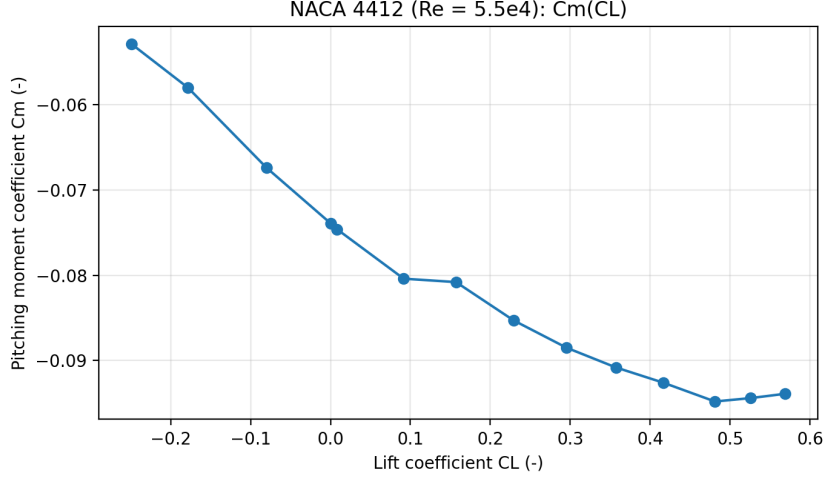


Figure 5: Moment coefficient C_m vs. lift coefficient C_L for NACA 4412 at $Re \approx 5.5 \times 10^4$, used to determine aerodynamic center location.

where the pitching moment remains constant with varying angle of attack. When the CG is ahead of this point, any nose-up disturbance creates a restoring moment that pitches the aircraft back down, ensuring stability. Conversely, if the CG were behind the aerodynamic center, disturbances would be amplified rather than damped, leading to instability.

Therefore, the FAA recommendation of placing the CG at $\sim 20\%$ MAC is indeed a good zone: it ensures the CG remains well ahead of the aerodynamic center ($20\% < 25.9\%$), providing positive static stability with sufficient margin. The distance between CG and aerodynamic center defines the *static margin*, a key stability parameter.

In our case, we were not seeking maximum stability: the objective was to maintain a sufficiently stable aircraft, but more maneuverable (more responsive pitch response) to allow for the perching maneuver. Perching requires aggressive pitch-up authority to achieve high angles of attack during the terminal approach phase. A more aft CG reduces static stability and brings the aircraft closer to its stability limit, thereby improving pitch maneuverability at the cost of reduced stability margins. The FAA notes that an increasingly aft CG makes stall recovery progressively more difficult, which clearly illustrates the loss of stability associated with it Federal Aviation Administration.

Simulations showed that a CG position at 40% of the MAC works satisfactorily for our perching configuration, providing sufficient stability while maintaining the necessary pitch maneuverability. Using the cambered-wing aerodynamic center estimate ($x_{ac}/c \approx 0.259$), this chosen CG position ($x_{cg}/c \approx 0.40$) places the CG *aft* of the wing aerodynamic center. This configuration is typical for conventional tail aircraft: while the CG is behind the wing's AC, the horizontal tail provides a stabilizing contribution that shifts the overall *neutral point* (the AC of the complete aircraft) further aft. A stable aircraft requires x_{cg} to remain ahead of this overall neutral point; the selected tail volume coefficient $V_H \approx 0.5$ ensures a positive static margin while retaining sufficient control authority for the perching maneuver. The resulting 40% MAC constraint directly influenced the choice of lightweight construction materials and the mass distribution strategy, particularly for the modular

wings which must remain lightweight to avoid shifting the CG adversely when swapping wing geometries.

3.2 Material Selection

The wings and tail could not be too heavy, as this would compromise both the total mass and the CG position. A first option considered was 3D printing; however, even with a low thickness (≈ 1 mm), the resulting mass remained too high.

We therefore sought a material that was:

- Lightweight (compatible with the 150 g constraint),
- Sufficiently rigid/strong for a cambered wing,
- Machinable with available means.

XPS foam (extruded polystyrene) met these requirements and was selected.

3.3 Manufacturing Process and Machining Strategy

Several machining processes were possible, notably hot wire cutting, but this method was less precise and more difficult to make reliable for an asymmetric and cambered geometry. We therefore chose CNC machining, which is better suited to ensure repeatability and profile precision.

However, machining remained non-trivial due to the need to machine both faces while maintaining precise repositioning. The machining strategy was defined using the Manufacturing workspace in Autodesk Fusion 360 and exported as G-code for the CNC machine. The process utilized a 6 mm flat-end milling cutter for both the reference holes and the surface machining.

The specific parameters used for both the wing and elevator were:

- **Toolpath:** Parallel bidirectional passes along the wing span.
- **Step-over:** 3 mm.
- **Tolerance:** 0.1 mm.

The two-sided machining strategy proceeded as follows:

1. **Reference Setup:** Two reference holes were drilled into the stock material using the 6 mm cutter to establish the coordinate system.
2. **First Face:** The first side was machined with a "bottom height" limit, ensuring the cutter did not penetrate through the entire stock thickness. This kept the part attached to the raw block.
3. **Flipping:** The block was flipped and positioned onto pins inserted into the reference holes to maintain alignment.

4. **Second Face:** The second side was machined, this time allowing the cutter to go to the full depth to detach the part from the stock.

This method aimed to ensure the alignment of both faces without losing geometric references during flipping.

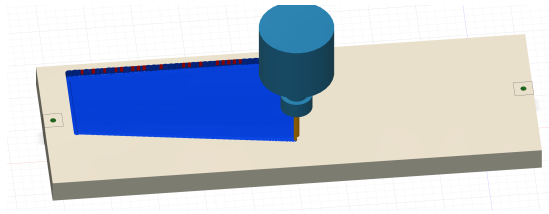


Figure 6: Preparation of the cutting strategy.

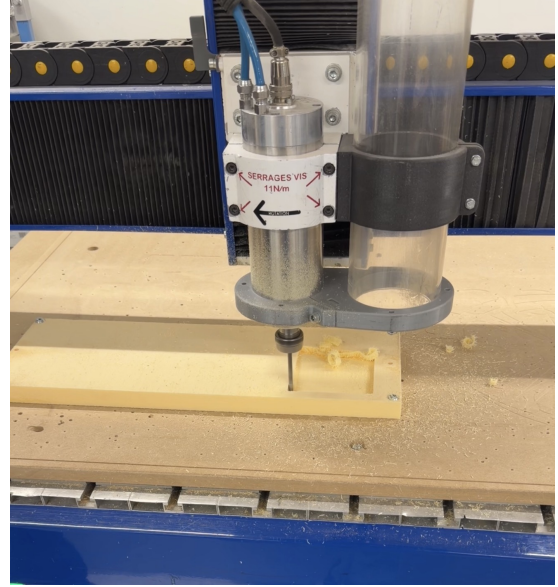


Figure 7: CNC machine cutting the wing.

3.4 CNC Machining Issues

During the machining of the XPS foam wings, several defects appeared:

Misalignment between the two faces: Despite repositioning, the machine (or the flipping operation) was slightly offset, which created a mismatch between the two passes and thus an imperfect final geometry.

Excessively rough finish: The step-over and/or finishing strategy were too "aggressive," making the tool path marks very visible on the surface.

Fragile thin sections: Some very thin parts of the wing did not hold mechanically; the foam cracked or broke during machining or handling.

3.5 Final Assembly

After manufacturing the elevator (Figure 8) using the same technique and cutting the fuselage (Figure 11), the final assembly was carried out as follows:

Overall reinforcement: Addition of carbon rods/tubes to stiffen the structure.

Removable wings: As emphasized in the introduction, the modular wing design is central to this project's objective of serving as a testbed for shape-optimization algorithms. The fuselage-wing connection (Figure 9) is made via a 3D-printed element (Figure 10), combined with two-component epoxy glue (hot glue was not retained as it was judged

insufficient in mechanical strength). The 3D-printed part is inserted inside the fuselage to improve assembly stability (better guidance and reduced play) and ensure mechanical wing attachment. The assembly is locked by a screw-nut system, allowing removable mounting and rapid wing replacement. This design enables systematic comparison of different airfoil geometries while maintaining a consistent fuselage, control system, and overall aircraft configuration.

Elevator hinge: The elevator is connected to the fuselage by a carbon rod (axis). Problem observed: without guidance, the elevator exhibits parasitic movements. Solution to apply: add washers/spacers to lock the elevator and impose a single degree of freedom (pure rotation).

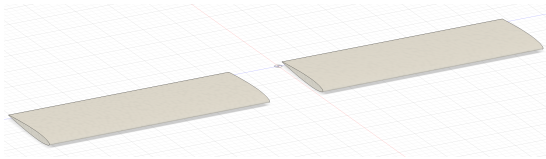


Figure 8: elevator

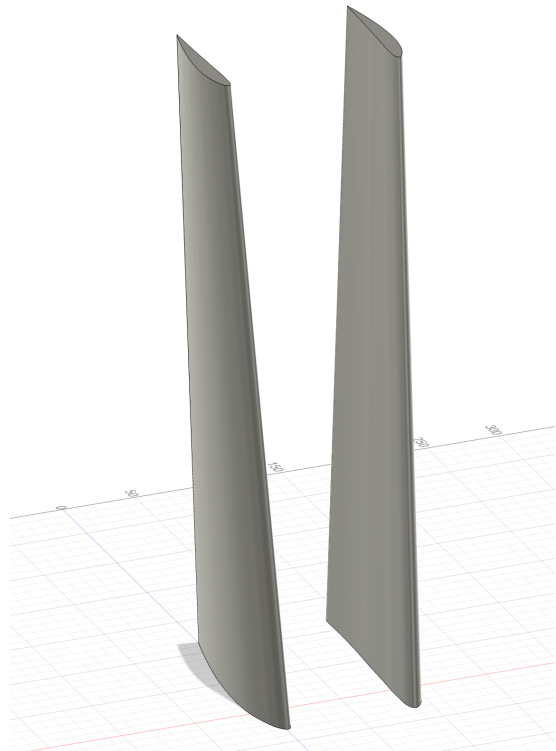


Figure 9: wings

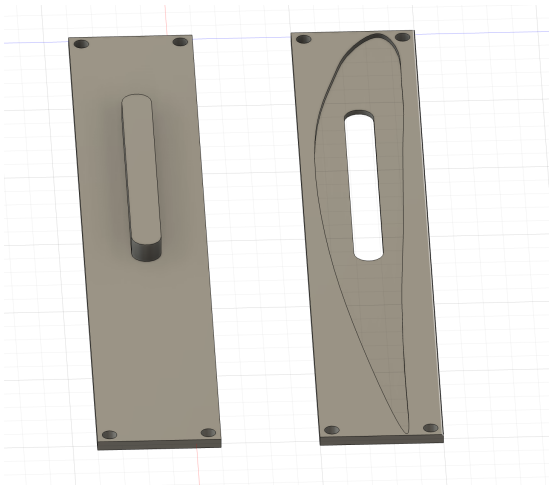


Figure 10: 3D-printed connection part.

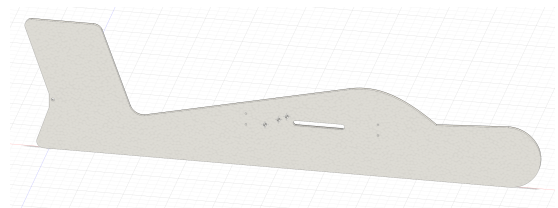


Figure 11: Fuselage

4 Control System

4.1 General Architecture

The control system is based on two Arduino microcontrollers: an Arduino Uno and an Arduino Nano. The Arduino Uno serves as the main controller, managing flight control logic and receiving commands from ground control. The Arduino Nano acts as a secondary controller, dedicated to specific tasks such as sensor data acquisition and servo control. This dual-microcontroller architecture provides modularity and allows independent management of critical subsystems.

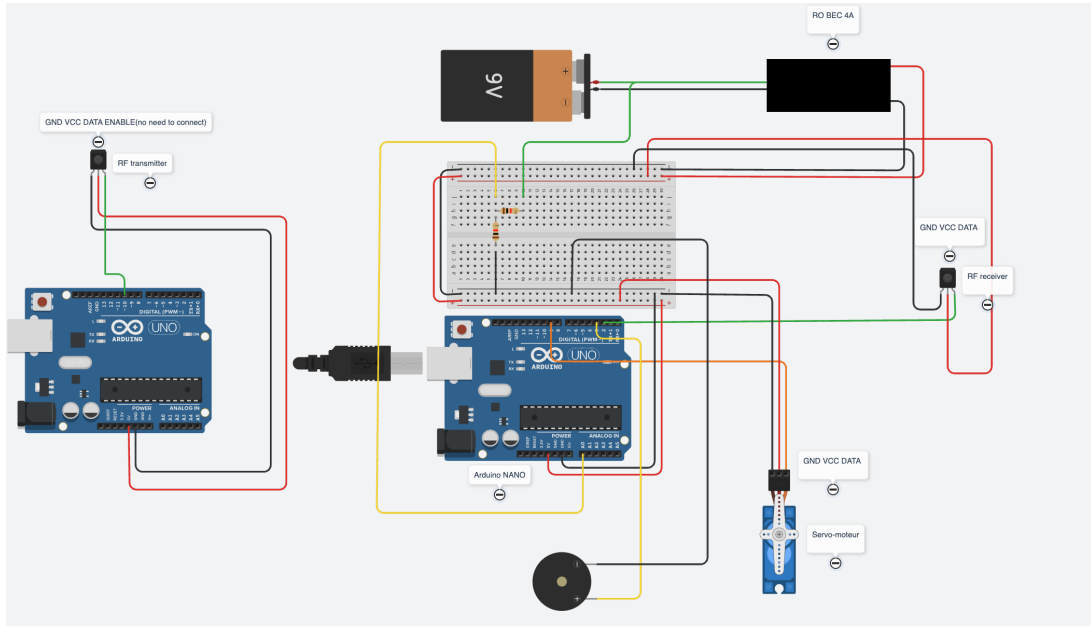


Figure 12: Control system circuit diagram showing the dual-microcontroller architecture, power distribution, and connections.

4.2 Power Supply

The entire system is powered by a 2S Li-Po battery, which provides a nominal voltage ranging from 8.4 V (fully charged) to 7.4 V (nominal). To power the microcontrollers and servos, which require a stable 5 V supply, a BEC (Battery Elimination Circuit) rated at 4 A is used. The BEC regulates the battery voltage down to 5 V, ensuring a stable power supply regardless of battery discharge level.

4.3 Battery Voltage Monitoring

To prevent deep discharge of the Li-Po battery, which could permanently damage it, a voltage monitoring system has been implemented. The battery voltage is measured via the analog input A0 of the Arduino. Since the Arduino's analog inputs accept a maximum

of 5 V, while the battery can reach up to 8.4 V, a voltage divider circuit is used to scale down the battery voltage to a safe range for the microcontroller.

The voltage divider consists of two resistors that reduce the battery voltage by a known ratio, allowing the Arduino to read a proportional voltage within its 0–5 V range. The microcontroller continuously monitors this voltage and calculates the actual battery voltage using the divider ratio.

An under-voltage protection threshold is set at 7.0 V (corresponding to a deeply discharged 2S Li-Po). If the monitored voltage falls below this threshold, the Arduino triggers an audible alarm via a buzzer, warning the operator to land the aircraft immediately and recharge the battery before further use. This protection mechanism prevents battery damage and ensures safe operation.

4.4 433 MHz Radio Communication and Protocol Robustness

For command transmission, we used a 433 MHz radio frequency receiver. The software implementation relies on the RCSwitch library Özgür, which is essential in this context as it contributes to signal detection, partial noise rejection, and extraction of received bits.

Several problems specific to this type of RF link were encountered and addressed:

Duplicate receptions / “bouncing” (message persistence at the receiver):

In our setup, one transmitted command frame (8-bit payload) is not necessarily decoded only once. After a single transmission, the RF receiver may keep triggering reception events for the same decoded value over a short period (≈ 300 ms). In our measurements, a single command can be reported up to 4 times: one reception at $t = 0$ (expected), followed by additional duplicate callbacks carrying exactly the same payload within the next ~ 300 ms.

This becomes problematic when commands are transmitted faster than this persistence window (i.e., with an inter-command period < 300 ms). In that case, duplicates of an older command can be interleaved with newer ones, producing an apparent out-of-order sequence at the receiver, e.g., $\text{cmd1} \rightarrow \text{cmd2} \rightarrow \text{cmd1} \rightarrow \text{cmd3}$, even though cmd1 was sent only once. Without filtering, this would cause the same actuator command to be applied multiple times and may also revert temporarily to a previous command.

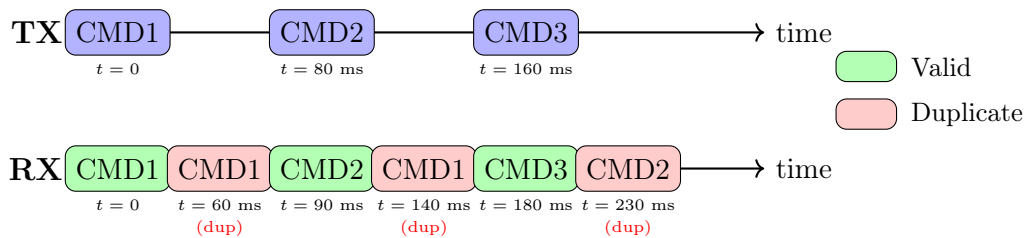


Figure 13: Example of duplicate receptions: the receiver may re-trigger the same decoded frame for ~ 300 ms, which can interleave with newer commands when the command period is < 300 ms.

To make the communication robust against these repetitions, the transmitted message

was defined as 16 bits, structured as follows:

- 8 bits for a counter (incremental message identifier),
- 8 bits of data corresponding to the angle command (useful range 0 to 120° for the servo motor).

On the receiving side, a message is only accepted if its counter differs from the last counter already processed. However, a simple “different counter” check is insufficient: delayed duplicates from very old commands could have counters that differ from the current one but should still be rejected. To address this, we implement a *security window* with wraparound-aware logic:

```
int difference = (int)counter - (int)lastCounter;
if (difference < 0) {
    difference += 256; // Wraparound correction
}

// Security window: reject messages that are too old
if (difference > 105) {
    mySwitch.resetAvailable();
    return; // Old message ignored
}
```

Security window logic:

- *Acceptance zone* ($\text{difference} \in [1, 105]$): Messages with a counter 1 to 105 units ahead are accepted. This allows up to 105 newer messages without issue.
- *Rejection zone* ($\text{difference} > 105$): Messages with a counter apparently far ahead (> 105) are rejected. These are actually delayed duplicates of old commands. For example, if `lastCounter` = 10 and we receive `counter` = 5:
 - Raw difference: $5 - 10 = -5$
 - After wraparound correction: $-5 + 256 = 251$
 - $251 > 105 \rightarrow$ Message rejected (it is an old message)
- *Minimum protection time*: If commands are sent rapidly (one new command every 10 ms), the window protects against duplicates for $105 \times 10 \text{ ms} = 1.05 \text{ s} > 300 \text{ ms}$ (the observed persistence duration).

Thus, even if duplicate frames are received multiple times, they are accepted only once, and old delayed duplicates are properly filtered out.

Intermittent frame corruption at high transmission rates: A second problem, observed recurrently but not completely eliminated, concerns data corruption when the command update rate becomes high. In our setup, a Python program sends commands

at approximately 100 Hz. However, low-cost 433 MHz ASK/OOK modules are specified for only a few kbps-class transfer rate (e.g., “Transfer rate: 4KB/s” in typical module documentation Robu.in), and they provide no built-in framing, CRC, or acknowledgments. In addition, when using the RCSwitch library, each “message” is not just 16 raw bits: it is transmitted as a pulse train including a relatively long synchronization pattern and a pulse-encoded representation of each bit (e.g., for common protocol definitions: pulse length $\approx 350 \mu\text{s}$, sync pattern {1,31}, and bit encoding {1,3}/{3,1}) Arduino Forum. Therefore, at high command rates (≈ 10 ms between commands), transmissions can overlap in time (especially when repeated transmissions are enabled), leaving insufficient guard time for the receiver and decoder Mbed OS. In practice this can lead to partial frame corruption: the counter field may still decode correctly while some bits of the angle field are flipped or mis-timed, resulting in a correct counter but an incorrect angle value Arduino Forum.

Interference between RF interrupts and servo PWM generation: The third major problem is related to the fact that RF reception generates many interrupts on the microcontroller side.

Why the RF receiver is connected to an interrupt pin: The 433 MHz receiver module does not output bytes directly: it outputs an asynchronous digital waveform (OOK/ASK) whose information is encoded in the timing of HIGH/LOW pulses. To decode this signal reliably, the receiver DATA pin is connected to an external interrupt pin (e.g., Arduino Uno D2/INT0) so that every rising/falling edge triggers an ISR, where we timestamp the edge (μs resolution) and reconstruct the bitstream (as done by the RCSwitch library). A polling approach is possible in theory, but it risks missing edges under CPU load and gives less robust decoding.

How RF interrupts disturb software-based servo PWM: During reception, the DATA line generates many interrupts over short time windows (one interrupt per edge). With classic “software PWM” servo control (e.g., Servo library), the servo pulse timing depends on timer interrupts and interrupt latency. When RF decoding ISRs occur frequently (and sometimes briefly block other interrupts), the servo PWM update can be delayed: the HIGH pulse can remain active longer than intended, producing visible jitter/spikes on the servo.

Mitigation: hardware-timer pulse generation: To make the servo command independent of RF interrupt load, we moved servo pulse generation to hardware timer output compare (microcontroller timer driving the pin). With hardware output compare, the pin transitions are generated by the timer peripheral rather than by software ISR scheduling, so the servo pulse width remains stable even when RF reception generates a high interrupt rate.

5 Results and Discussion

5.1 Comparison with Reference Platform

Our prototype differs significantly from the glider developed by Moore et al. [2014]. While Moore's design focused on minimizing weight (≈ 100 g) and maximizing control authority using a simple flat-plate wing and an all-moving tail, our approach prioritized modularity and aerodynamic experimentation. By targeting a 150 g mass budget to accommodate a modular, cambered wing (NACA 4412), we created a more aerodynamically efficient platform in cruise, but one with higher inertia and wing loading.

The choice of a cambered airfoil successfully demonstrated that complex shapes could be manufactured and flown, validating our CNC workflow. However, the increased mass and the conventional elevator design resulted in a faster, less maneuverable aircraft compared to the "floaty" and agile reference glider. Where Moore's glider could easily enter a deep stall, our prototype behaves more like a conventional aircraft, favoring stable forward flight over the extreme post-stall maneuvers required for perching.

5.2 Suitability for the Perching Task

In the context of the specific perching task, the current design has shown mixed results. On the one hand, the project succeeded in delivering a reliable, flyable platform with a robust control and communication architecture. The modularity objective was met, allowing for wing swapping as intended.

On the other hand, the flight tests indicate that the current configuration is not yet capable of autonomous perching. The primary limiting factors are the lack of pitch authority (insufficient to pitch up rapidly into a stall) and the relatively high sink rate due to wing loading. While the glider is stable, it lacks the "explosive" maneuverability needed to transition from glide to perch in the short distance available. The design is thus a successful *experimental testbed* for aerodynamic research, but requires the mechanical and aerodynamic refinements detailed in the following section to become a successful *percher*.

6 Future Work and Recommendations

While the current prototype successfully demonstrates stable flight and a robust control architecture, the ultimate goal of autonomous perching remains to be achieved. The flight tests revealed that the glider currently lacks the "floatiness" and aggressive pitch authority required to execute the high-angle-of-attack terminal maneuver. Based on these observations, we propose the following improvements for the next design iteration.

6.1 Enhancing "Perchability": Aerodynamics and Mechanics

Wing Loading and Area: The current glider, weighing 150 g with a wing area of 0.10 m^2 , has a wing loading of approximately 1.5 kg/m^2 . While flyable, the aircraft feels heavy in the air and sinks relatively fast, reducing the time available for the controller to stabilize the approach. To improve the "floaty" character of the glider and reduce the stall speed, we recommend increasing the wing area in the next iteration. A lower wing loading would provide a larger safety margin for the transition to post-stall flight.

Pitch Authority and Elevator Design: The most critical limitation observed is the inability to pitch up aggressively enough to enter the deep-stall regime required for perching. This is likely due to two factors:

- *Elevator Range of Motion:* The current mechanical linkage limits the elevator deflection to a standard range ($\approx \pm 20 - 30^\circ$). In contrast, Moore et al. [2014] utilized a tail with a massive range of motion ($> 80^\circ$) to maintain control authority in separated flow. Future designs must redesign the fuselage rear section and the elevator hinge mechanism to physically allow for these extreme deflections without mechanical interference.
- *Tail Volume:* While the current tail volume ($V_H \approx 0.5$) is sufficient for stability, it may be insufficient for the violent control moments needed for perching. Increasing the horizontal tail area (S_h) would directly increase the pitching moment available to initiate the maneuver.

Center of Gravity (CG): In practice, we could not achieve exactly 40% MAC, but a lower value. This parameter must be adjusted and tested iteratively to optimize for the best position. While moving it aft could theoretically increase pitch responsiveness by reducing the static stability margin, this is a delicate trade-off: an overly aft CG could make the glide phase uncontrollable. We suggest prioritizing the increase in tail authority (area and deflection) alongside these CG adjustments.

6.2 Next-Generation Avionics: Weight and Connectivity

The current electronics stack (Arduino Uno + Nano + RF Receiver + Custom PCB) is robust but heavy and complex. A significant weight reduction and capability upgrade could be achieved by modernizing the avionics.

Transition to WiFi (Arduino Nano 33 IoT): We propose replacing the dual-microcontroller setup with a single **Arduino Nano 33 IoT**. This board integrates a powerful microcontroller and a WiFi module in a compact form factor.

- *Simplified Communication:* Instead of the error-prone 433 MHz OOK protocol that required complex custom framing and "security windows," the command link could use standard WiFi protocols (UDP or TCP). This would inherently handle packet integrity, framing, and error checking, eliminating the "duplicate reception" and "frame corruption" issues described in Section 4.

- *Direct Actuation:* The Nano 33 IoT has sufficient I/O and timers to generate servo PWM signals directly, removing the need for a secondary slave controller.

Weight Reduction and Telemetry: Adopting the Nano 33 IoT would allow for the removal of the Arduino Uno, the external RF receiver module, the buzzer, and potentially the custom PCB. This could save 20-30 g, directly contributing to the wing loading reduction mentioned above. Furthermore, safety features like the low-voltage alarm could be improved: instead of a heavy buzzer, the glider could transmit real-time battery voltage telemetry to the ground station laptop via WiFi, providing the pilot with better situational awareness.

6.3 Manufacturing Optimization

To address the geometric imperfections observed in the current prototype (specifically warping due to machining vibrations), the manufacturing process must be refined.

1) Reduce misalignment (alignment/flipping): Improve the referencing strategy during flipping to limit the offset between the two machined faces and ensure better geometric consistency.

2) Improve surface finish: Add a dedicated finishing pass with:

- Lower step-over,
- Adapted feed rate,
- Smaller tool (or better suited to the material),
- Reduced depth of cut.

Provide for light sanding (or application of a thin "skin" such as lightweight film/tape) if necessary, while remaining compatible with the mass constraint.

3) Avoid breakage and vibration-induced warping: The strategy of fully detaching the part during the second pass caused significant vibrations, resulting in a final geometry that was not perfectly straight. To address this:

- **Use Holding Tabs:** Instead of fully detaching the wing with the CNC, leave a continuous thickness of 1–2 mm or specific tabs around the edges to maintain structural rigidity during the final cuts.
- **Manual Post-Processing:** These tabs can then be easily cut manually with a cutter after machining is complete. This would significantly reduce vibration and improve the geometric accuracy of the trailing edge and thin sections.
- Locally thicken critical areas in the CAD model to increase mechanical strength.
- Consider using balsa wood, paying particular attention to mass, as any weight increase directly impacts the dynamic behavior and stability of the glider.

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